

# AI - BASED PHISHING DETECTION FRAMEWORK

## PRESENTED BY

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# INTRODUCTION

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Phishing is a cyberattack technique in which attackers create fraudulent websites that imitate trusted organizations to steal confidential information from users. With the increasing number of online transactions and digital services, phishing attacks have become more frequent and sophisticated.

Traditional phishing detection methods depend on manually updated blacklists and predefined rules, making them ineffective against newly generated phishing websites. Artificial Intelligence and Machine Learning provide an efficient alternative by learning patterns from datasets and identifying malicious URLs based on extracted features.

This project develops an AI-powered phishing detection framework capable of automatically classifying URLs into legitimate or phishing categories using Machine Learning algorithms.

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## PROBLEM STATEMENT

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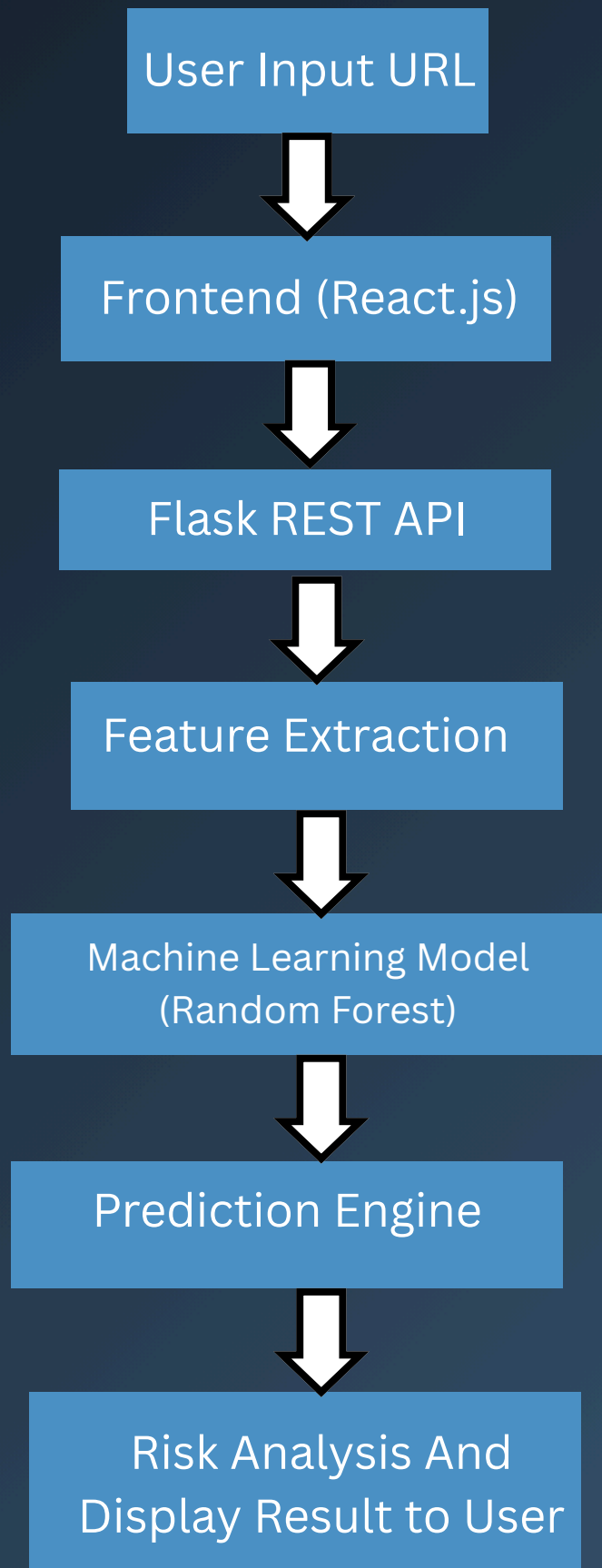
Existing phishing detection techniques mainly rely on blacklist databases and manual rule-based systems, which are unable to detect newly generated phishing websites effectively. This limitation increases the risk of cyberattacks and data theft. Therefore, there is a need for an intelligent, scalable, and automated phishing detection framework capable of accurately identifying phishing websites in real time using Machine Learning.

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# SYSTEM ARCHITECTURE

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# TECHNOLOGY USED

| Technology    | Purpose                |
|---------------|------------------------|
| Python        | Backend Development    |
| Flask         | REST API Development   |
| React.js      | Frontend Development   |
| HTML          | User Interface         |
| CSS           | Styling                |
| JavaScript    | Frontend Logic         |
| Pandas        | Data Processing        |
| NumPy         | Numerical Computation  |
| Scikit-learn  | Machine Learning       |
| Random Forest | URL Classification     |
| Git           | Version Control        |
| GitHub        | Source Code Management |
| Vercel        | Frontend Deployment    |
| Render        | Backend Deployment     |

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# DATASET

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The project uses a combined dataset containing phishing and legitimate website URLs collected from publicly available cybersecurity datasets. Data preprocessing was performed to remove inconsistencies, handle missing values, and prepare the dataset for Machine Learning.

Dataset preprocessing included:

- Removing duplicate records
  - Handling missing values
  - Feature extraction
  - Data cleaning
  - Label encoding
  - Train-test splitting
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# FEATURE EXTRACTION

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The following URL-based features were extracted:

- Having IP Address
- Presence of '@' Symbol
- URL Length
- URL Path
- Prefix/Suffix Detection
- HTTPS Protocol
- Tiny URL Detection
- Redirection Symbols
- Number of Subdomains
- Domain Age
- Website Traffic
- DNS Record
- URL Depth
- Domain Registration Information

These features enable the Machine Learning model to identify suspicious URL characteristics effectively.

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# METHODOLOGY

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**The project follows the following workflow:**

**Step 1:** Collect phishing and legitimate URL datasets.

**Step 2:** Clean and preprocess the collected data.

**Step 3:** Extract URL-based security features.

**Step 4:** Train the Random Forest Machine Learning model.

**Step 5:** Save the trained model.

**Step 6:** Develop Flask REST APIs.

**Step 7:** Design React frontend.

**Step 8:** Integrate frontend and backend.

**Step 9:** Deploy the application.

**Step 10:** Perform testing and evaluation.

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# MACHINE LEARNING MODEL

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The Random Forest Classifier was selected because it provides high prediction accuracy, reduces overfitting, supports multiple input features, and performs well on classification problems.

## Model Training Process:

- Data preprocessing
- Feature extraction
- Train-test split
- Model training
- Model evaluation
- Model serialization
- Prediction

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# FRONTEND IMPLEMENTATION

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The frontend was developed using React.js to provide a simple and interactive user interface.

## Major Features:

- URL input field
  - Scan button
  - Risk prediction display
  - Loading indicator
  - User-friendly interface
  - Responsive design
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# BACKEND IMPLEMENTATION

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The backend was developed using Flask and provides REST APIs for communication between the frontend and Machine Learning model.

Main APIs:

GET /

Returns backend status.

POST /predict

Accepts user URL, extracts features, predicts phishing probability, and returns the result.

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## DEPLOYMENT

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The project was successfully deployed using cloud platforms.

Frontend Deployment

- Vercel

Backend Deployment

- Render

Source Code Repository

- GitHub

Deployment enables users to access the phishing detection system from anywhere through a web browser.

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# TESTING

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Testing was performed to verify the accuracy and functionality of the developed system.

Testing Types:

- Functional Testing
- API Testing
- User Interface Testing
- Machine Learning Prediction Testing
- Integration Testing

## Sample Result:

Input URL

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Feature Extraction

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Prediction

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Risk Analysis

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Displayed Result

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# RESULTS

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The developed framework successfully classifies URLs into phishing and legitimate categories.

Achievements:

- High prediction accuracy
- Fast response time
- Real-time URL analysis
- Smooth frontend-backend integration
- Successful cloud deployment
- Reliable phishing detection

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# CHALLENGES FACED

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- Dataset preprocessing and cleaning
  - Feature extraction consistency
  - Model integration with Flask
  - API communication issues
  - Frontend-backend connectivity
  - Deployment configuration
  - Version control using Git and GitHub
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# FUTURE SCOPE

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**The project can be further enhanced by implementing:**

- Deep Learning-based phishing detection
- Browser extension integration
- Email phishing detection
- QR code phishing detection
- Mobile application
- Real-time threat intelligence
- Cloud-based monitoring dashboard
- Continuous model retraining

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# CONCLUSION

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The AI-Based Phishing Detection Framework successfully demonstrates the use of Machine Learning for identifying phishing websites. The developed system combines URL feature extraction, Random Forest classification, a React-based frontend, and a Flask backend to provide accurate and real-time phishing detection. The project improved practical knowledge of Machine Learning, Web Development, REST API integration, cloud deployment, and collaborative software development. The developed framework provides an effective and scalable cybersecurity solution that can be extended for real-world phishing detection applications.